

PROJECT REPORT

HR Analytics Dashboard

Employee Attrition & Workforce Insights

Developed using Microsoft Power BI

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Table of Contents

No.	Section
1	Project Overview
2	Dataset and Metrics
3	Methodology
4	Dashboard Overview
5	Results and Analysis
6	Key Insights
7	Business Recommendations
8	Limitations and Future Scope
9	Conclusion

Note: fill in the personal details on the cover page before submission.

1. Project Overview

This project is an interactive HR Analytics Dashboard built in Microsoft Power BI. Its objective is to analyze employee information, workforce trends and employee attrition, helping HR teams see **who is leaving, from where and why**. The dashboard compares attrition across departments, salary ranges, job roles, age groups, gender and experience.

Headline KPIs

31	25	6	19.4%	34.74	7.61
Total Employees	Active Employees	Attrition Count	Attrition Rate	Average Age	Avg Experience (years)

2. Dataset and Metrics

The dataset holds **31 employee records**. The dashboard view analyzed is filtered to the **Marketing** department and the **26-35** age group.

- **Departments available:** Administration, Finance, Human Resources, Marketing, Operations, Sales.
- **Job roles in view:** Sales Executive, Sales Representative.
- **Fields analyzed:** attrition (Yes/No), department, salary slab, job role, job satisfaction (1-4), age group, gender, age and experience.

Metric definitions

Metric	Meaning / Formula
Active Employees	Total minus attrition count (31 - 6 = 25)
Attrition Rate	Attrition count / Total employees = 6 / 31 = 19.4%
Average Age	Mean age of employees: 34.74 years
Avg Experience	Mean total work experience: 7.61 years

3. Methodology

- **Step 1 - Data preparation:** load the employee dataset into Power BI, check data types, and clean blanks and duplicates.
- **Step 2 - DAX measures:** create measures for total employees, active employees, attrition count and rate, average age and average experience.
- **Step 3 - Visualization:** build cards, donut chart, bar chart, matrix, column chart, pie chart and area chart.
- **Step 4 - Interactivity:** add Age Group and Department slicers that filter every visual together.

4. Dashboard Overview

The single-page dashboard has a title banner, six KPI cards, a Department slicer, an Age Group slicer and seven visuals.

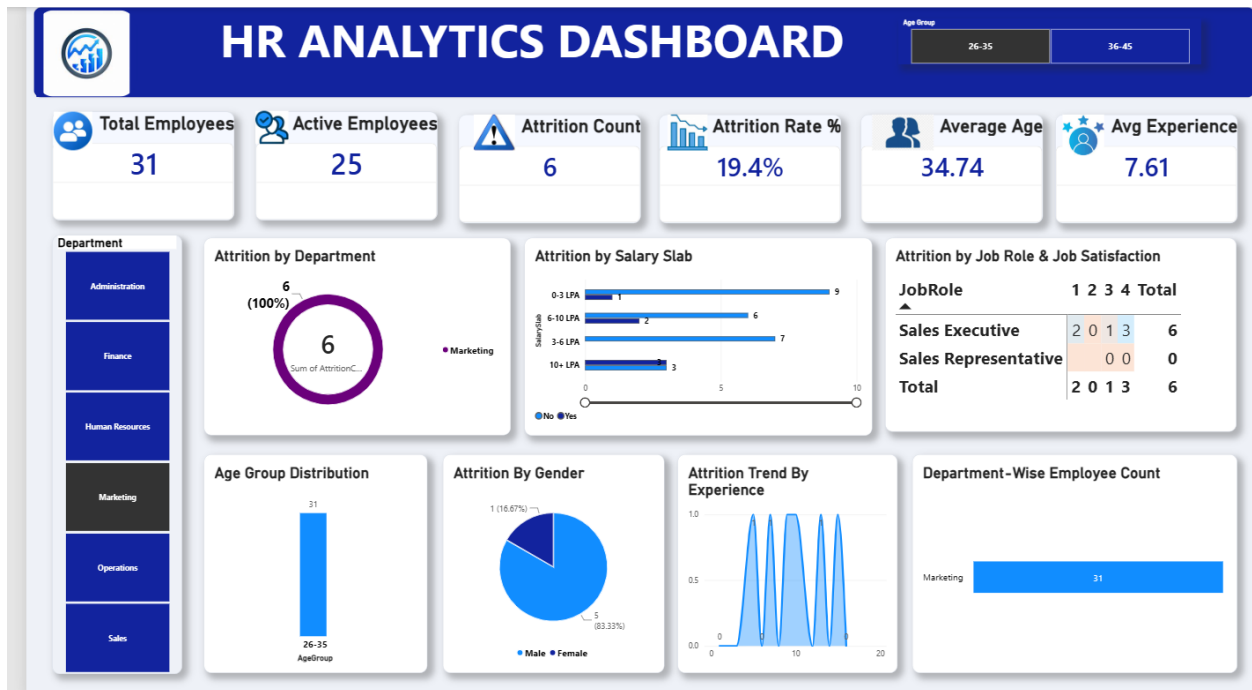


Figure 1: HR Analytics Dashboard (Power BI)

Visuals included

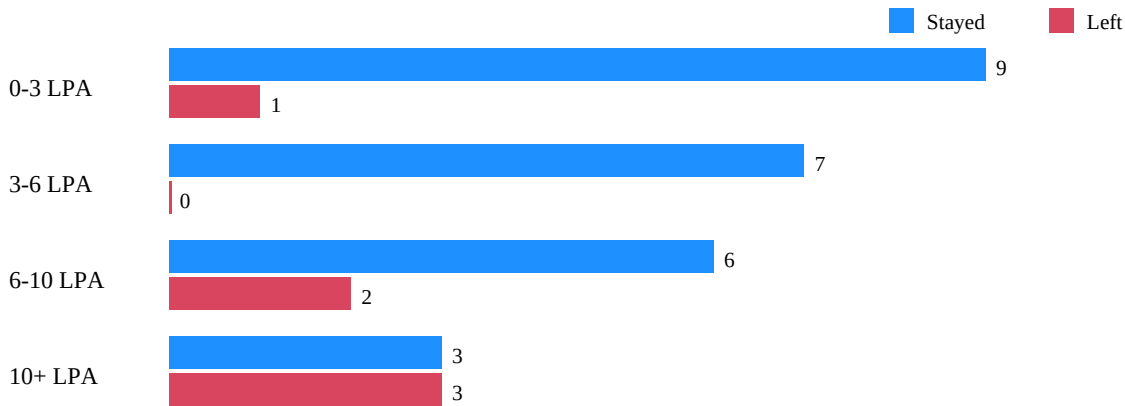
Visual	Purpose
Attrition by Department	Shows how exits are split across departments
Attrition by Salary Slab	Compares stayed vs left across pay bands
Job Role & Job Satisfaction	Matrix of satisfaction scores (1-4) by role
Age Group Distribution	Employee count by age band
Attrition by Gender	Gender-wise split of leavers
Attrition Trend by Experience	Exits across years of experience
Department-wise Employee Count	Headcount per department

5. Results and Analysis

5.1 Attrition by Department

All 6 exits (100%) belong to Marketing. Department-wise count also shows 31 employees in Marketing under the applied filter.

5.2 Attrition by Salary Slab

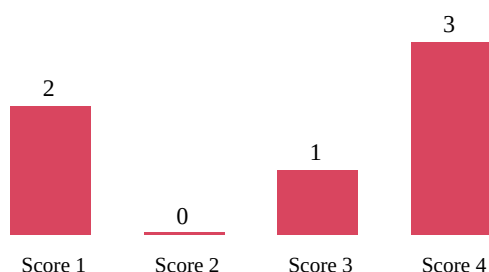


Salary slab	Stayed	Left	Attrition rate
0-3 LPA	9	1	10%
3-6 LPA	7	0	0%
6-10 LPA	6	2	25%
10+ LPA	3	3	50%
Total	25	6	19.4%

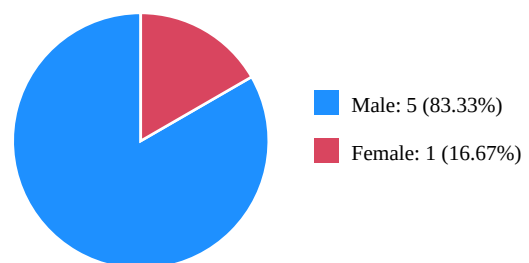
Figure 2: Attrition by salary slab. The 10+ LPA band has the highest rate.

5.3 Job Role and Job Satisfaction

All 6 leavers are Sales Executives (satisfaction 1: two, 2: none, 3: one, 4: three). Sales Representatives show zero attrition.



Satisfaction scores of leavers



Attrition by gender

5.4 Age, Gender and Experience

- **Age:** all 31 records fall in the 26-35 age group.

- **Gender:** 5 of 6 leavers are male (83.33%) and 1 is female (16.67%).
- **Experience:** leavers appear at several experience levels up to about 16 years, in separate clusters rather than one steady rise.

6. Key Insights

- **Attrition is 19.4%** - about 1 in 5 employees (6 of 31) has left.
- **Higher pay bands lose more people:** 10+ LPA has a 50% rate and 6-10 LPA has 25%, while 3-6 LPA lost nobody.
- **Satisfied employees still left:** 3 of 6 leavers rated satisfaction 4, so satisfaction alone does not explain exits.
- **Exits are concentrated** in the Sales Executive role within Marketing.
- **Most leavers are male** (5 of 6); gender totals are needed to confirm whether the rate is actually higher.

7. Business Recommendations

- **Review pay and growth** for 6+ LPA Sales Executives with career paths, promotions and retention bonuses.
- **Run stay and exit interviews** to learn about manager support, targets and growth, since satisfaction scores did not explain exits.
- **Support mid-career staff** through mentoring and role changes where exits cluster.
- **Monitor monthly:** refresh the dashboard every month and set an alert if attrition passes a set threshold (for example 15%).

8. Limitations and Future Scope

Limitations

- Small sample (31 records), so percentages change easily.
- Only one department and one age group are in view.
- No exit-reason or performance data is available.

Future scope

- Add all departments and more records.
- Predict attrition risk using machine learning.
- Add tenure, promotion and manager data.
- Publish to Power BI Service with scheduled refresh.

9. Conclusion

The HR Analytics Dashboard gives HR teams a clear, filterable view of workforce attrition. In this dataset 19.4% of employees left, mostly Sales Executives in Marketing, with higher-paid staff leaving at the highest rate. These findings support targeted retention actions and better workforce planning.